Data Cleaning

2024 general election data by SV precinct

```
## load 2024 general election vote counts by precinct
library(tidyverse)
g24raw <- read_csv("data/g24_sov_by_g24_svprec.csv", na = c("", "NA", "***"))

## focus on important cols
g24 <- g24raw |>
  select(COUNTY, FIPS, SVPREC, CDDIST,
         CNGDEM01, CNGDEM02, CNGIND01, CNGREP01, CNGREP02)

## isolate all rows with a congressional district of 0
g24_cddist0 <- g24 |>
  filter(CDDIST == 0)
# They appear to be aggregations of other rows and have precinct names that
#  start with `SOV` or end in `TOT`.

## remove those rows
g24 <- g24 |>
  filter(CDDIST != 0)

## inspect rows with NA for counts
all_na <- g24 |>
  filter(is.na(CNGDEM01))
all_na
# they seem innocuous. i'll retain them.

## check for replicates
g24 |>
  count(SVPREC) |>
  ggplot(aes(x = n)) +
  geom_bar()
# yes, there are replicates!

# find the worst offender
g24 |>
  count(SVPREC) |>
  arrange(desc(n))

# look at it
```

```
g24 |>
  filter(SVPREC == "01")
# zomg, SVPREC isn't a unique id. each county does their own numbering.
# election data is very de-centralized / unsystematic.

## check that County x SVPREV can serve as a unique id
g24 |>
  group_by(COUNTY, SVPREC) |>
  summarize(n = n()) |>
  arrange(desc(n))
# thank god

write_csv(g24, "data/g24-clean.csv")
```

Re-running the 2024 election results under the new district map

Approach A
Load and clean voting data by SR Precinct.

```
library(tidyverse)
sr <- read_csv("data/state_g24_sov_data_by_g24_srprec.csv", na = c("", "NA",
"***"))

sr <- sr |>
  select(COUNTY, FIPS, SRPREC, SRPREC_KEY, CDDIST, CNGDEM01,
         CNGDEM02, CNGIND01, CNGREP01, CNGREP02) |>
  filter(CDDIST != 0)
```

Load SR precinct shapefiles and append voting data.

```
library(sf)
sr_shp <- read_sf("data/shapefiles/srprec_state_g24_v01_shp.shp")

# correct errors in geometries
sr_shp <- sr_shp |>
  st_transform(3310) |>          # switch to equal area projection first
  st_set_precision(1) |>         # snap points to 1 m grid
  st_make_valid() |>             # fix self-intersections/bow-ties
  st_collection_extract("POLYGON")

sr_shp <- sr_shp |>
  left_join(sr, by = "SRPREC_KEY")
```

Load new district shapefiles and perform interpolation.

```
d25 <- read_sf("data/shapefiles/AB604.shp") |>
  st_transform(crs = 3310)

approach_A <- sr_shp[c("CNGDEM01", "CNGDEM02", "CNGREP01", "CNGREP02")] |>
  st_interpolate_aw(d25, extensive = TRUE, na.rm = TRUE)
```

```
approach_A$dist <- d25$DISTRICT

approach_A <- approach_A |>
  mutate(d_votes = CNGDEM01 + CNGDEM02,
         r_votes = CNGREP01 + CNGREP02)

write_csv(approach_A, "data/g24-rerun-approachA.csv")
```

Approach B

Load and clean voting data by SR Precinct.

```
library(tidyverse)
sr <- read_csv("data/state_g24_sov_data_by_g24_srprec.csv", na = c("", "NA",
"***"))

sr <- sr |>
  select(COUNTY, FIPS, SRPREC, SRPREC_KEY, CDDIST, CNGDEM01,
         CNGDEM02, CNGIND01, CNGREP01, CNGREP02) |>
  filter(CDDIST != 0)
```

Load block file and join with vote data by SR precinct.

`BLKTOTREG`: The total number of registered voters on that block. `PCTSRPREC`: The proportion of total registered voters in the precinct that live on that block.

```
blk <- read_csv("data/state_g24_sr_blk_map.csv")

blk <- blk |>
  left_join(sr)
```

Allocation precinct votes proportionally to (fractional) blocks based on proportion of registered voters.

```
blk <- blk |>
  mutate(dem1 = CNGDEM01 * PCTSRPREC / 100,
         dem2 = CNGDEM02 * PCTSRPREC / 100,
```

```
         rep1 = CNGREP01 * PCTSRPREC / 100,
         rep2 = CNGREP02 * PCTSRPREC / 100,)
```

Load block to AB 604 district correspondence and attach new districts to voting data.

```
blk_cd <- read_csv("data/cd25.csv", col_names = c("block_key", "new_district"))

blk <- blk |>
  left_join(blk_cd, by = join_by("BLOCK_KEY" == "block_key"))
```

Aggregate votes at new district level.

```
approach_B <- blk |>
  group_by(new_district) |>
  summarize(dem1 = sum(dem1, na.rm = TRUE),
            dem2 = sum(dem2, na.rm = TRUE),
            rep1 = sum(rep1, na.rm = TRUE),
            rep2 = sum(rep2, na.rm = TRUE)) |>
  mutate(d_votes = dem1 + dem2,
         r_votes = rep1 + rep2) |>
  drop_na(new_district)

write_csv(approach_B, "data/g24-rerun-approachB.csv")
```